

CHRONOS 2

TỪ DỰ BÁO ĐƠN BIẾN ĐẾN DỰ BÁO PHỔ QUÁT

Người trình bày: Bùi Hoàng Nhân
Applied Forecasting

Introduction & Methodology

Research Questions

- RQ1:** Can zero-shot foundation models match classical methods?
- RQ2:** Does fine-tuning improve foundation model performance?
- RQ3:** What are the performance vs. cost trade-offs?

Dataset & Approach

-  Bitcoin Wikipedia pageviews (2020-2024)
-  1,827 daily observations
-  30-day ahead forecasting
-  5-fold rolling-origin CV

Fine-Tuning Setup

-  25 epochs
-  Mixed precision (bfloat16)
-  4GB VRAM
-  ~30-50 min training

Category	Model	Key Feature
Baseline	Seasonal Naive	Last week's same day
Baseline	ETS	Exponential Smoothing
Baseline	Gradient Boosting	Engineered features
Foundation	Chronos-2 Zero-Shot	No training required
Foundation	Chronos-2 Fine-Tuned	Domain-adapted (NEW)

Key Results

Performance Rankings (MASE ↓ Lower is Better)			
Rank	Model	MASE	MAE
 1st	Gradient Boosting	0.344	873
 2nd	Chronos Fine-Tuned	0.373	946
 3rd	Chronos Zero-Shot	0.394	999
4th	Seasonal Naive	0.397	1007
5th	ETS	0.508	1288

Statistical Significance

MASE Improvement:	↓ 5.3%
sMAPE Improvement:	↓ 6.0%
Wilcoxon p-value:	0.028 ✓

Effect Metrics

Effect Size:	0.42 (medium)
Win Rate:	4/5 folds
Rank Change:	3rd → 2nd

Analysis & Insights

🏆 Gradient Boosting (MASE: 0.344) - Winner

- Explicit weekly pattern capture (lag_7: 35% feature importance)
- Domain-optimized features (lags [1,7,14,28], rolling means)
- Most consistent across folds (std=0.042)

🥈 Chronos Fine-Tuned (MASE: 0.373) - 2nd Place

- Domain adaptation learns Bitcoin-specific patterns
- Better calibration of prediction intervals
- Consistent improvement across 4/5 folds

🥉 Chronos Zero-Shot (MASE: 0.394) - 3rd Place

- Competitive without any training
- Generic architecture not optimized for weekly patterns
- Best use: Diverse time series portfolio

Error Analysis Summary

By Horizon: All models degrade $h=1 \rightarrow h=30$, GB maintains lead

By Level: High pageview spikes challenging (MASE > 0.60)

Consistency: GB most stable, ETS most variable

Performance vs. Cost Trade-offs

Model	MASE	Training Time	Best Use Case
Gradient Boosting	0.344	~20 sec	Single critical series
Chronos Fine-Tuned	0.373	~30-50 min	Domain-specific deployment
Chronos Zero-Shot	0.394	0 sec	Portfolio of diverse series

✓ Fine-tune if:

- 500+ training observations available
- Can afford ~30-50 min training
- Series has learnable patterns
- Want better calibrated forecasts

✗ Skip fine-tuning if:

- Very short series (<100 points)
- Need instant deployment
- Extreme accuracy required → use GB

💰 ROI Analysis

Investment: 30-50 min (one-time)

Gain: 5.3% improvement

Favorable ROI ✓

Practical Recommendations

1 Single Critical Series

e.g., sales forecasting

Choose: Gradient Boosting
MASE: 0.344 | **Time:** ~20 sec

2 Domain-Specific App

e.g., website traffic forecasting

Choose: Chronos-2 Fine-Tuned
MASE: 0.373 | **Time:** ~30-50 min

3 Portfolio of 100+ Series

e.g., product demand

Choose: Chronos-2 Zero-Shot
MASE: 0.394 | **Time:** 0 sec

4 Maximum Accuracy

mission-critical systems

Choose: Ensemble (GB + Chronos)
Reason: Complementary strengths

💡 Key Takeaway

Fine-tuning bridges the gap between zero-shot convenience and specialized model performance

Conclusions

Key Contributions

1. First comprehensive comparison of zero-shot vs. fine-tuned Chronos-2
2. Statistically validated 5.3% improvement ($p=0.028$)
3. Practical framework for performance-cost trade-offs
4. Open-source with memory-efficient fine-tuning (4GB VRAM)

Research Questions Answered

RQ1: Zero-shot competitive but not superior to GB (MASE: 0.394 vs 0.344, -13% gap)

RQ2: Fine-tuning yields **statistically significant 5.3% improvement**

RQ3: Fine-tuned offers favorable cost-performance balance

Final Recommendation

Fine-tune foundation models when: domain has 500+ observations, learnable patterns exist, and training time (~30-50 min) is acceptable. This represents the **optimal middle ground** between zero-shot and extensive feature engineering.

Future Work & Limitations

Current Limitations

Single domain (Wikipedia pageviews only)

Simplified fine-tuning framework

Univariate (no exogenous variables)

Prediction interval undercoverage (49% vs 80%)

Future Research Directions

Technical

True T5 gradient updates,
LoRA adapters, gradient
checkpointing

Evaluation

Multi-domain testing,
hierarchical forecasting,
exogenous variables

Production

Ensemble methods, online
learning, auto-tuning,
calibration

Expected improvement with true fine-tuning: 10-20% additional MASE reduction

Summary & Contact

Project Summary

Dataset: Bitcoin Wikipedia pageviews (1,827 days)

Models: 5 approaches (3 classical + 2 foundation)

Main Finding: Fine-tuning achieves 5.3% improvement

Rank: 2nd place ($p=0.028$)

Key Metrics

MASE Improvement: **5.3%**

Statistical Sig: **$p=0.028$ ✓**

Rank Change: **3rd → 2nd**

Training Time: **~30-50 min**

Reproducibility

github.com/Vipproplayerone1/ts-chronos-gpu

✓ Complete source code (12 modules)

✓ Documentation & README

✓ Configuration files

✓ All artifacts & visualizations

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